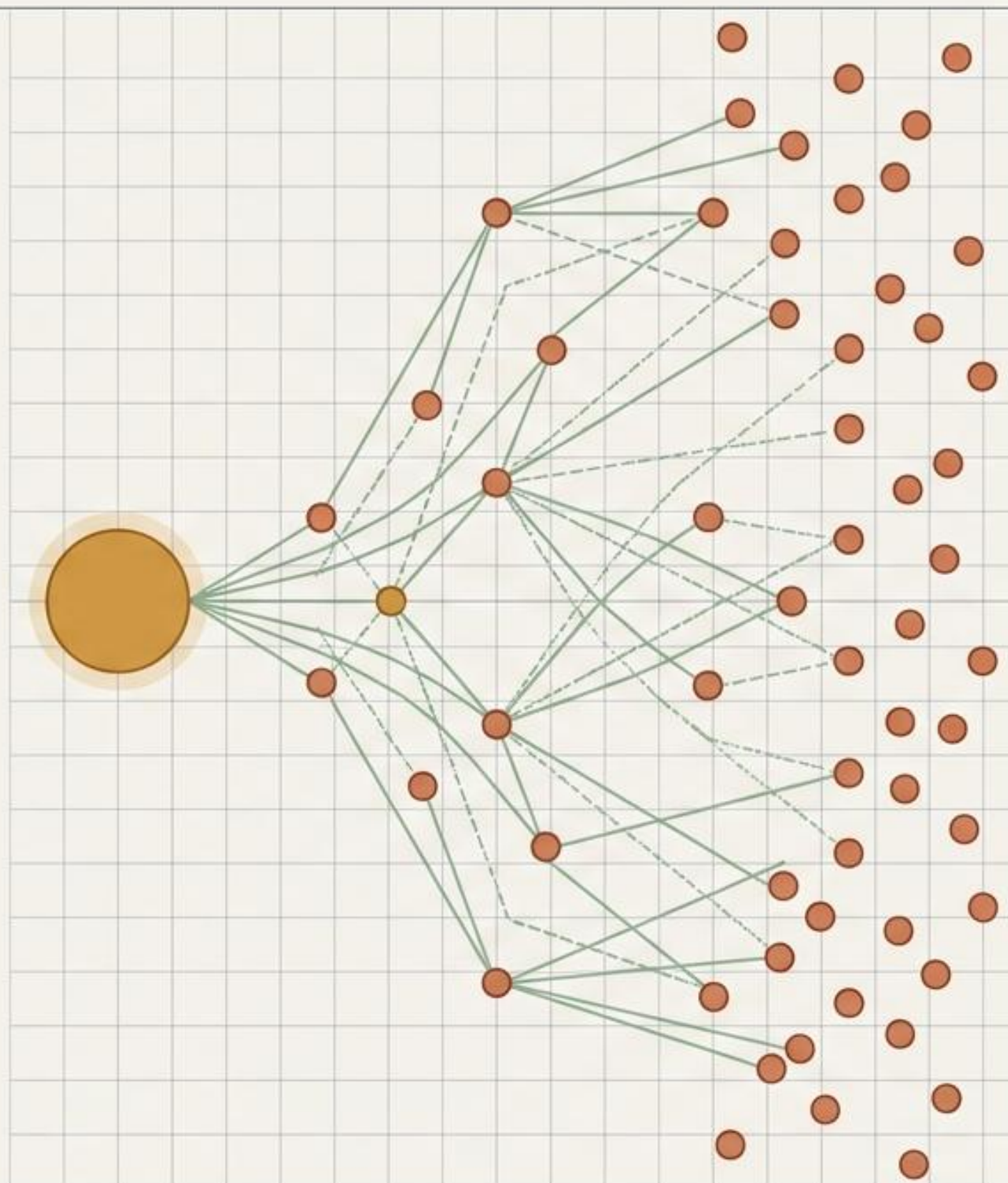


Identity Knowledge Graphs

Solving the Entity Resolution Crisis

dr Zbigniew Galar



The Human Nuance

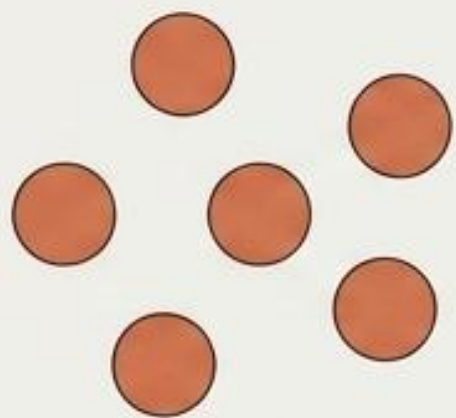
Jane Coleman marries and becomes Jane Downe. She opens a joint checking account using her passport, while her existing credit card uses her driver's license and maiden name.

The Relational Failure

1	Jane Coleman	Credit	Null Passport	License 49587640	DOB 1989-10-28
2	Jane Downe	Current	Null Passport	License RA14958	DOB 1989-10-28

No common alphanumeric identifiers.
A relational database sees two distinct people, leading to wasted marketing marketing spend and zero unified visibility.

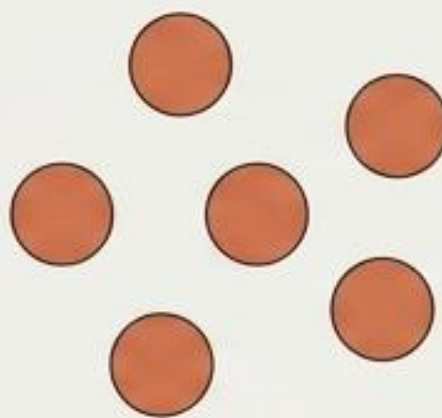
Where Duplication Hides



Data Integration

Siloed systems (Sales, CRM, Marketing).

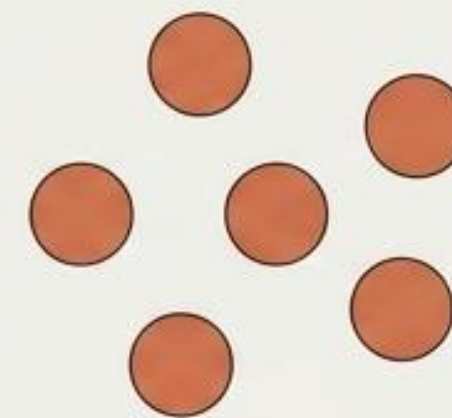
Records are identified and described differently by each department without consistent strong identifiers.



Anonymous Activity

Clickstreams and unauthenticated users.

Lack of logins forces reliance on behavioral traces like session times, cookie IDs, and navigational patterns.



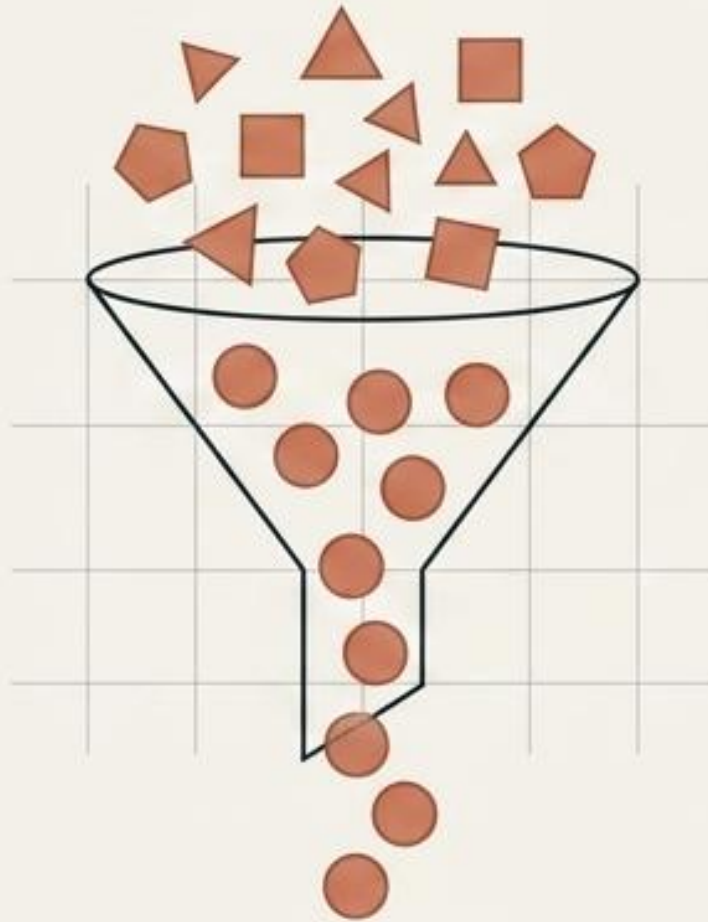
Intentional/Fraudulent Duplicates

Bad actors manipulating the system.

Individuals adjusting addresses for better quotes, abusing trials with fake emails, or sellers flooding catalogs.

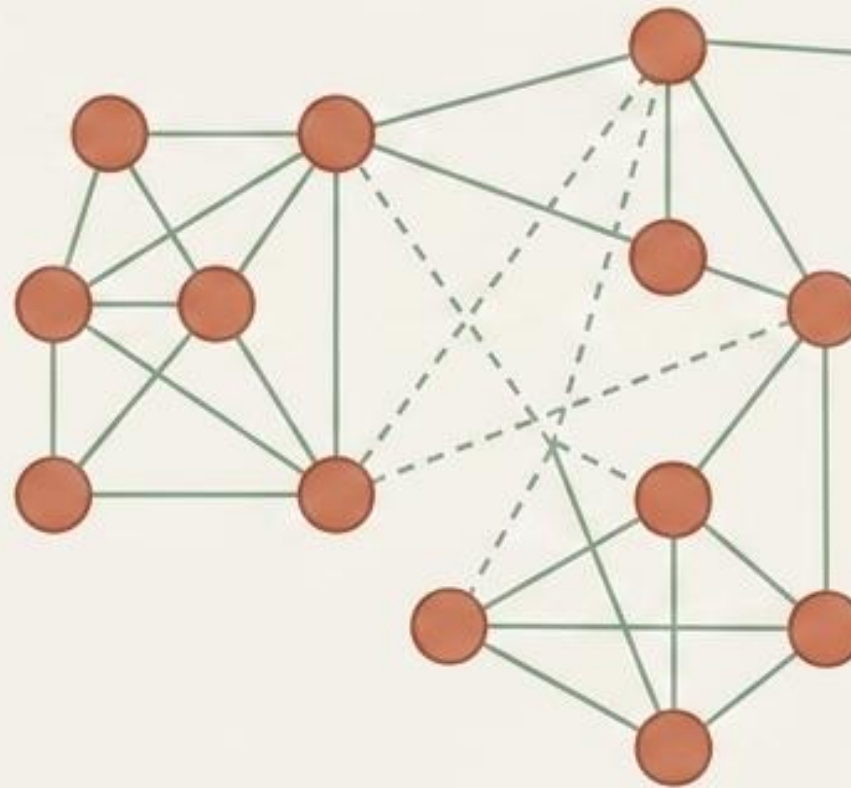
The Graph-Based Solution Architecture

1. Data Preparation



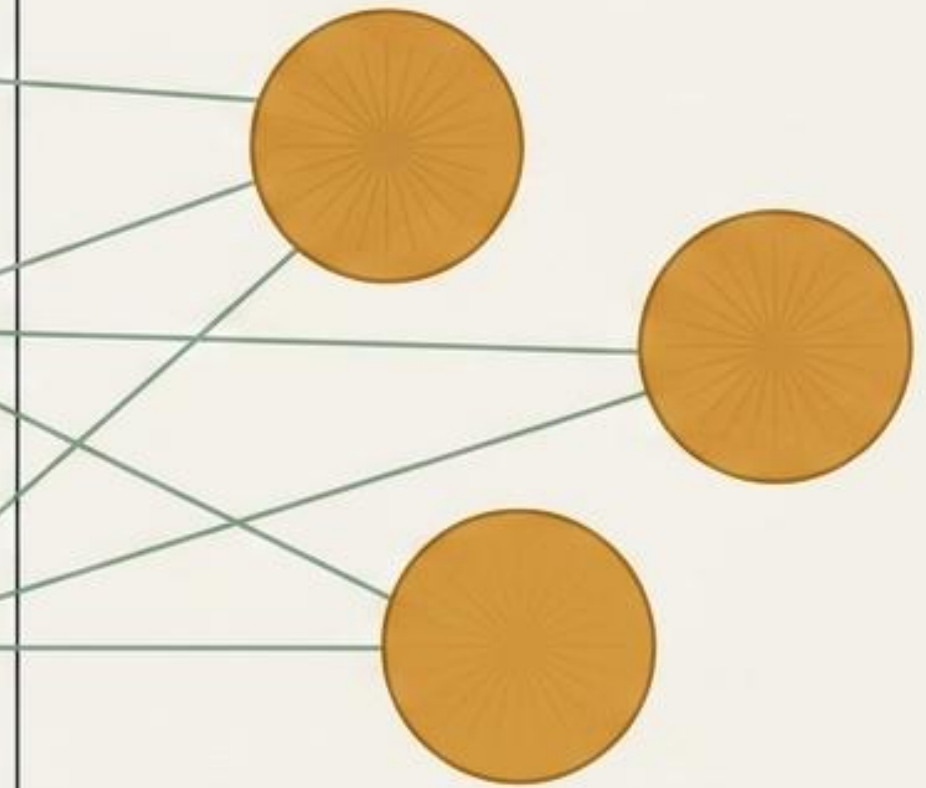
Harmonization &
Blocking Keys

2. Entity Matching



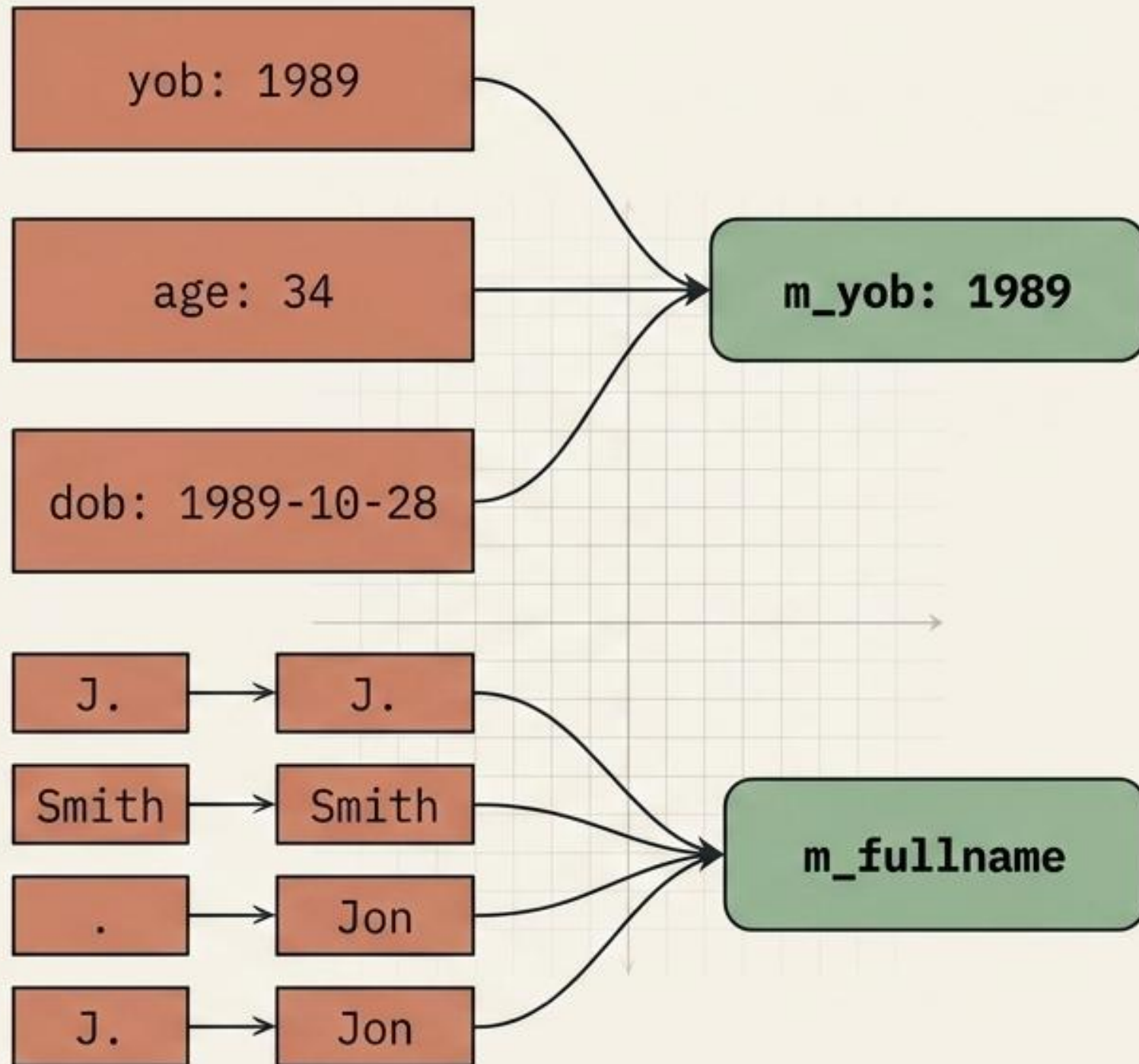
Algorithmic
Relationship Scoring

3. Master Entity Curation



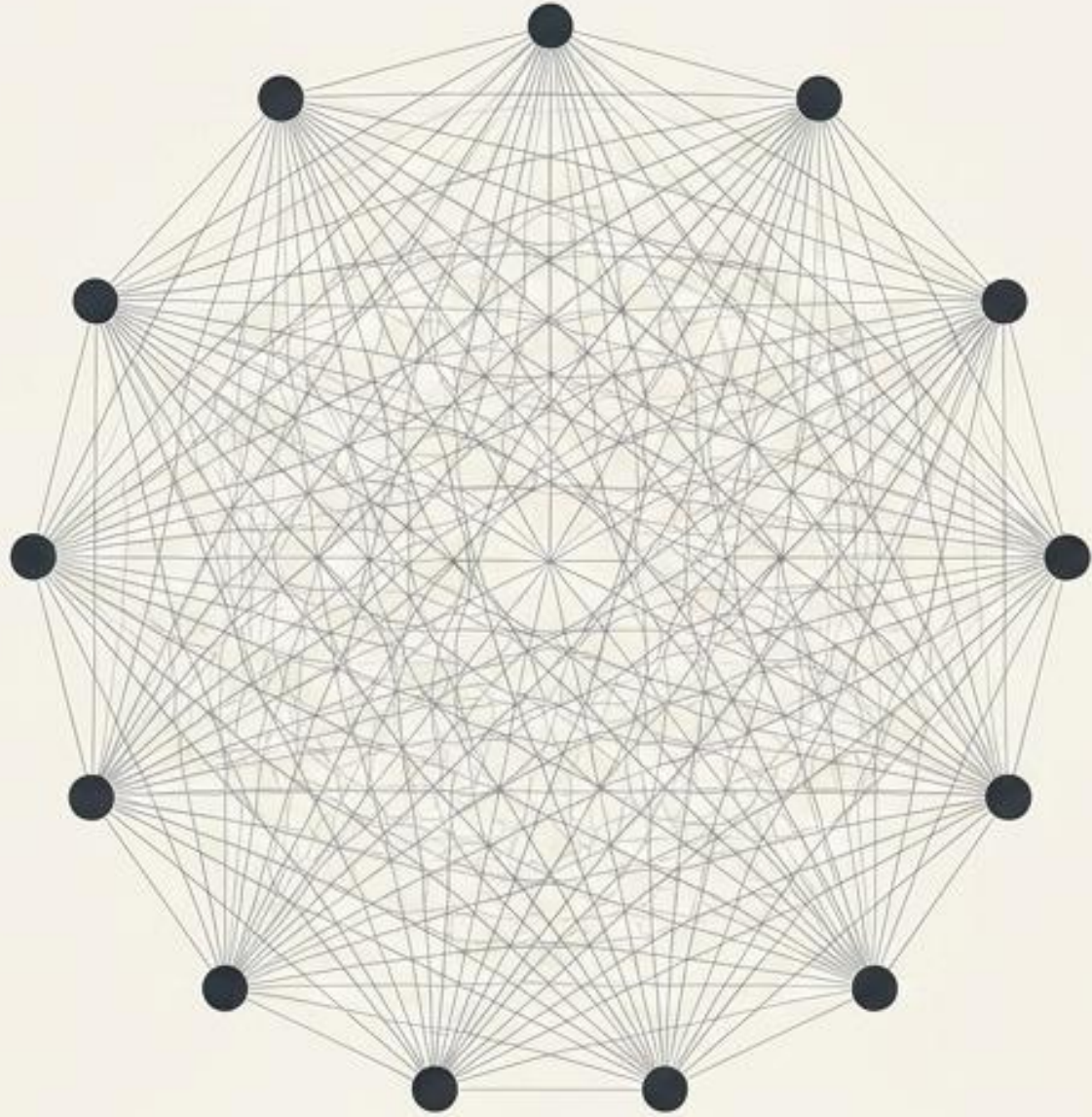
Weakly Connected
Components (WCC)

Step 1: Harmonization & Lowest Common Denominators

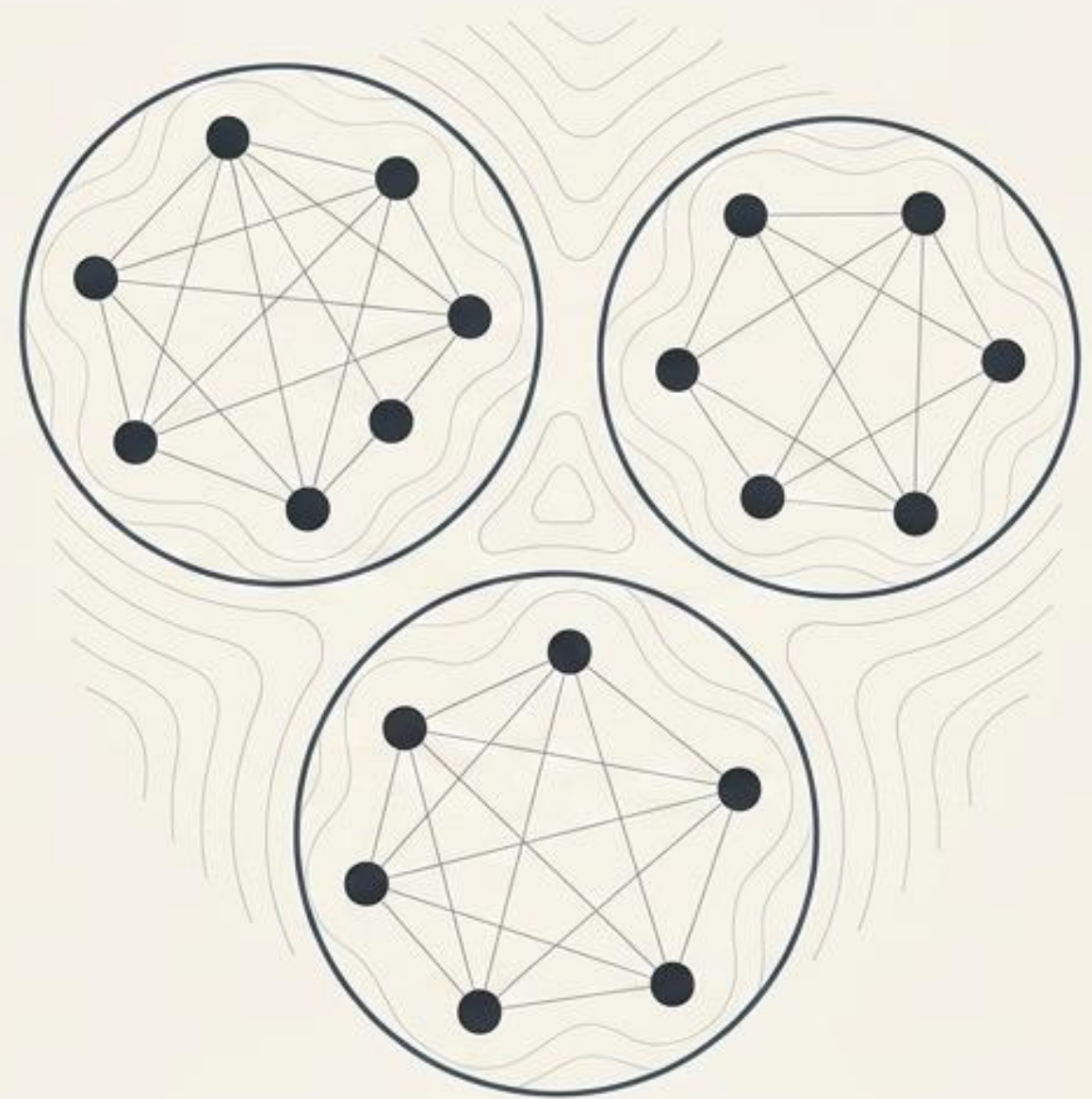


```
// Harmonizing variable age formats  
SET p.m_yob = date().year - toInteger(p.age)  
  
// Harmonizing complete birthdates  
SET p.m_yob = date(p.dob).year
```


The Scaling Bottleneck: Blocking Keys



Without Blocking: n^2 operations for n elements.



With Blocking Keys: Search space computationally reduced by isolating probable match pools.

The Feature Diagnostic Matrix

Feature Type	Examples	Resulting Graph Action
Strong Identifiers	Social Security Number (SSN), Passport Number, Product SKU	Exact matches generate an unambiguous <u>SAME_AS</u> relationship.
Weak Identifiers	Full Name, Physical Address, Email	High-probability similarities (<u>Jaro-Winkler distance < 0.2</u>) generate a <u>SIMILAR</u> relationship.
Non-Identifying Features	Year of Birth (m_yob), Gender, Timestamp	Used as corrective <u>weighting factors</u> to boost or dampen the <u>similarity score</u> .

Step 2: The 5-Step Matching Algorithm

Step 1: Match Strong IDs -> Create SAME_AS edge.

Step 2: Match Weak IDs above threshold -> Create weighted SIMILAR edge.

Step 3: Strong ID mismatch? -> Discard the SIMILAR edge.

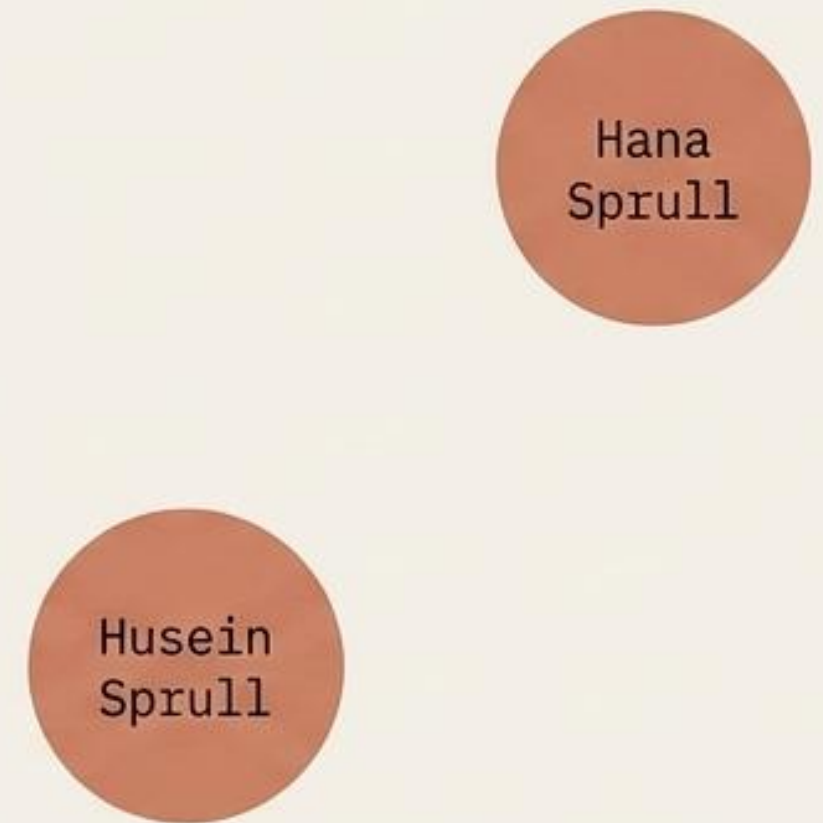
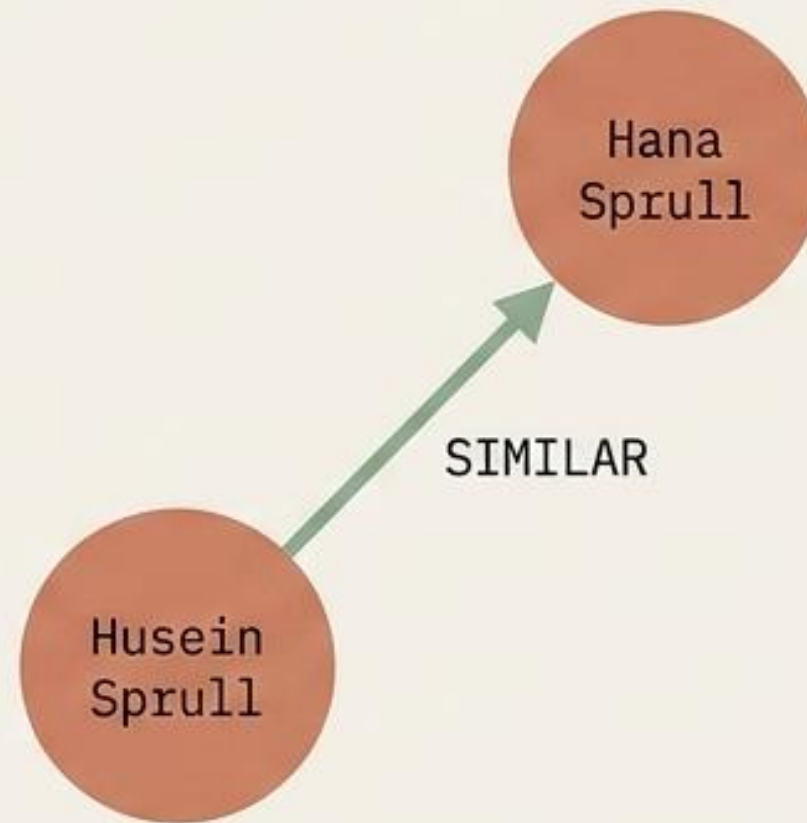
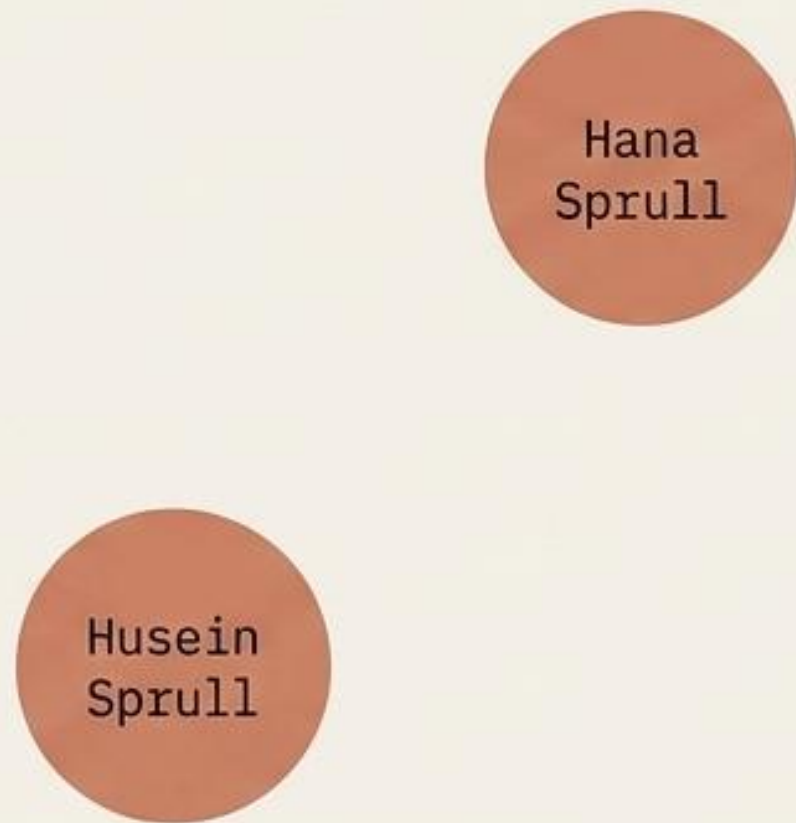


Step 4: Non-identifying feature match?
-> Apply correcting factor (boost/dampen weight).

Step 5: Score below minimum threshold?
-> Discard the SIMILAR edge.



Algorithmic Pruning: The Lifecycle of a Relationship

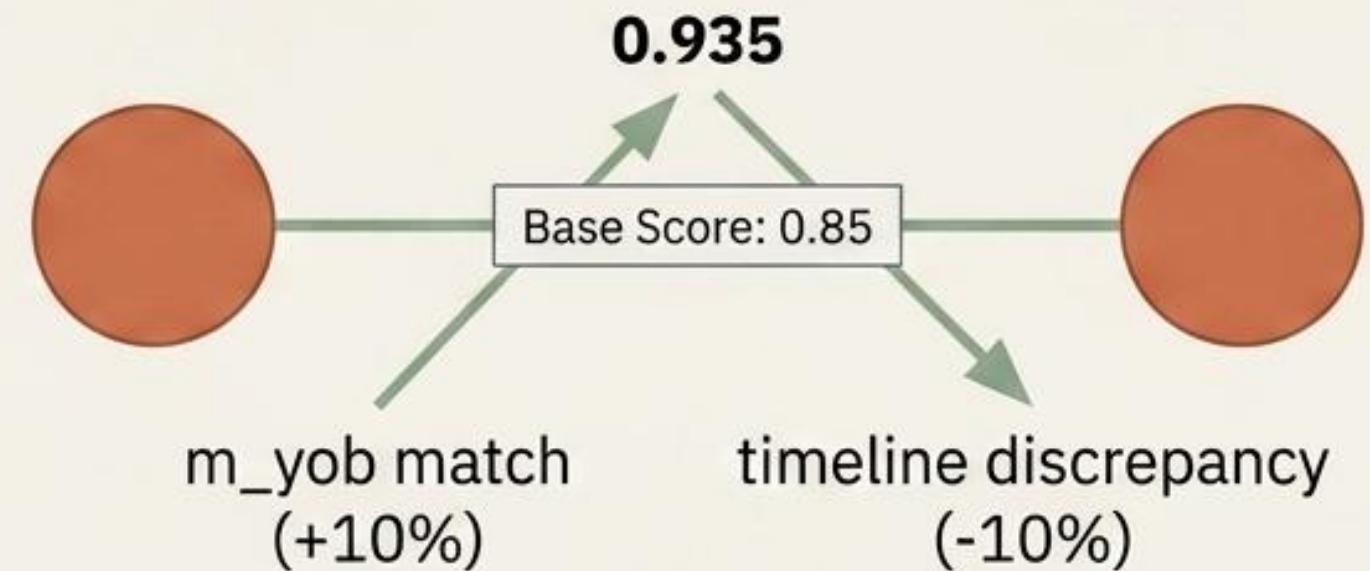


Jaro-Winkler string similarity on `m_fullname` detects a match distance < 0.2 . Edge created.

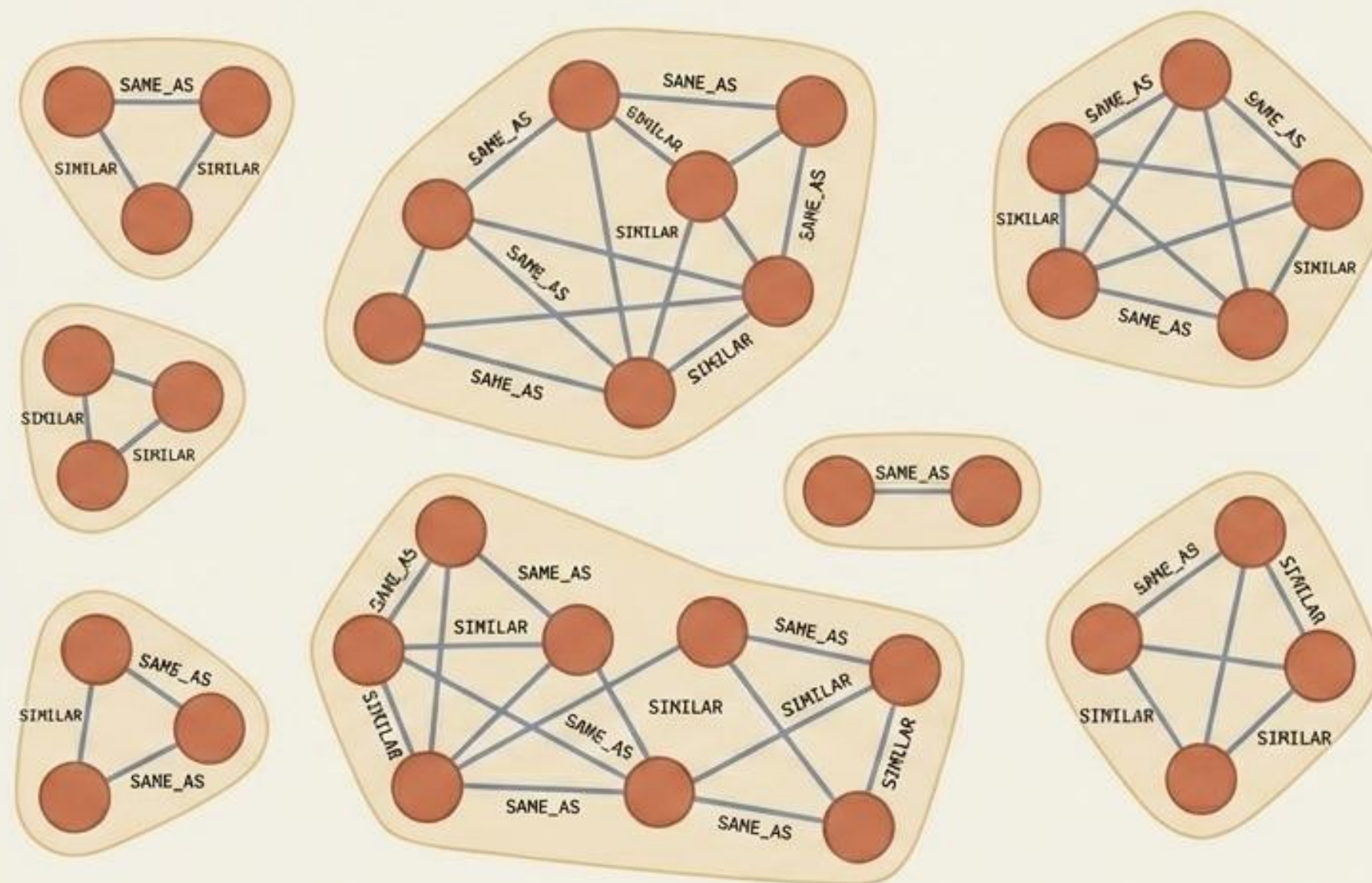
SSN mismatch overrides the weak identifier. Relationship destroyed to prevent false entity merger.

Applying Algorithmic Scoring (Boosting & Dampening)

```
SET sim.sim_score = sim.sim_score *  
  CASE WHEN yob_diff > $yob_threshold  
    THEN 0.9  
    ELSE 1.1 END
```

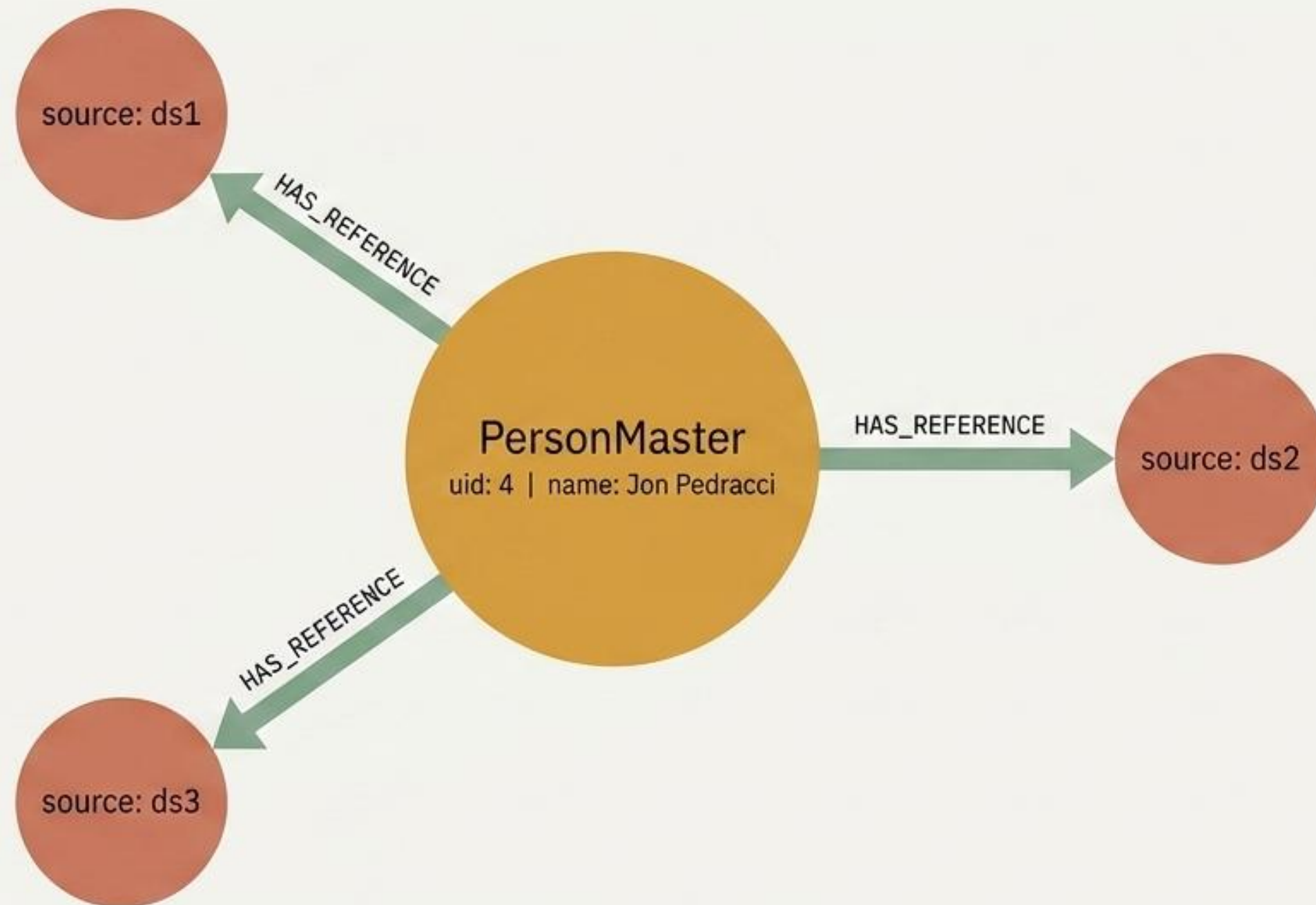


Step 3: Resolving Entities via Weakly Connected Components (WCC)



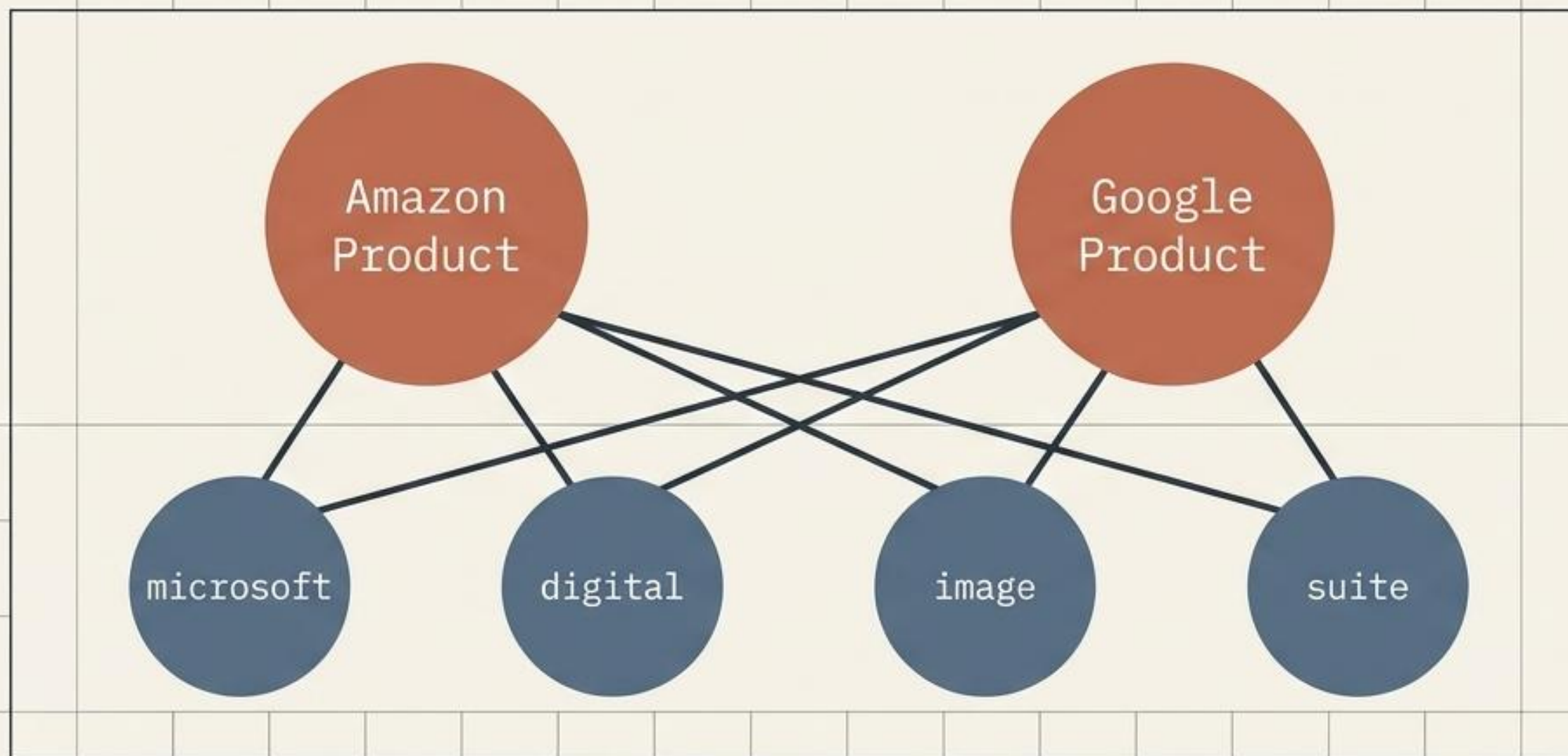
The WCC algorithm traverses undirected **SAME_AS** and **SIMILAR** relationships. Every discrete, enclosed subgraph it identifies represents a single, unique real-world entity

Persisting the Golden Record



```
{
  master_entity_id: 4,
  references: [
    { source: ds1, details: {...} },
    { source: ds2, details: {...} },
    { source: ds3, details: {...} }
  ]
}
```


Advanced Resolution: Unstructured Data & Tokenization



In the absence of identifiers, product descriptions are tokenized into Word nodes. The `gds.nodeSimilarity` algorithm computes pair-wise similarity using the Jaccard metric. Products sharing a high overlap of neighbor Word nodes are probabilistically merged.

Case Study: Meredith Corporation Identity Graph

Scale: 30+ consumer brands | 30 billion nodes | 67 billion properties | 4.4 Terabytes of data

Before: The Cookie Era

350 Million

Fragmented Profiles

14 Days

Touchpoint Lifespan

4

Average visits
per cookie

After: The Graph Era

163 Million

Rich, Consolidated Golden Records

241 Days

Touchpoint Lifespan

23.8

Average visits
per profile

A 20-30% deeper understanding of customers by tracking connections over time, transforming dark advertising into significant revenue gains.

The Virtuous Cycle of Master Data

